

FT85BD Manual V1.5

Specifications :

- Firmware: V1.5 or above (firmware update via Flipsky ESC Tool supported)
- Voltage: 14-85V (safe for 4S-20S)
- Continuous Current: 100A / Single; 200A / Dual (values depend on the mounting and ambient temperature around the device!)
- Peak Current: 250A / Single; 500A / Dual
- Communication Ports: USB, CAN, UART, PPM, ADC
- Input Signals Support: PPM, ADC, PAS, UART(compatible with Flipsky remote such as VX1, VX1Pro, VX2, VX3, VX3Pro, VX4Pro)
- Startup Modes Support: IPDS (sensorless), HFPI (sensorless), HALL (sensored),ENCODER(ABI Encoder , AS5047 Absolute Type Encoder SPI Port), OPEN(Sensorless Open-Loop Startup)
- Traction Control: YES
- BEC: 5V@1A
- Multiple Speed Gear levels: Low, Medium, High, Reverse
- Front Lights, Rear Brake Lights & Horn: Support
- Programmable: YES
- Regenerative Capacity: YES
- Slide Power on & Auto shutdown: Yes. Auto power off time can be customized.
- Self-resetting switch with green LED button Diameter: 16mm
- Power Wire Size: 10AWG; Motor Wire Size: 12 AWG
- Size: L132*W81*H26.5mm(Including Alu. Case)

Features:

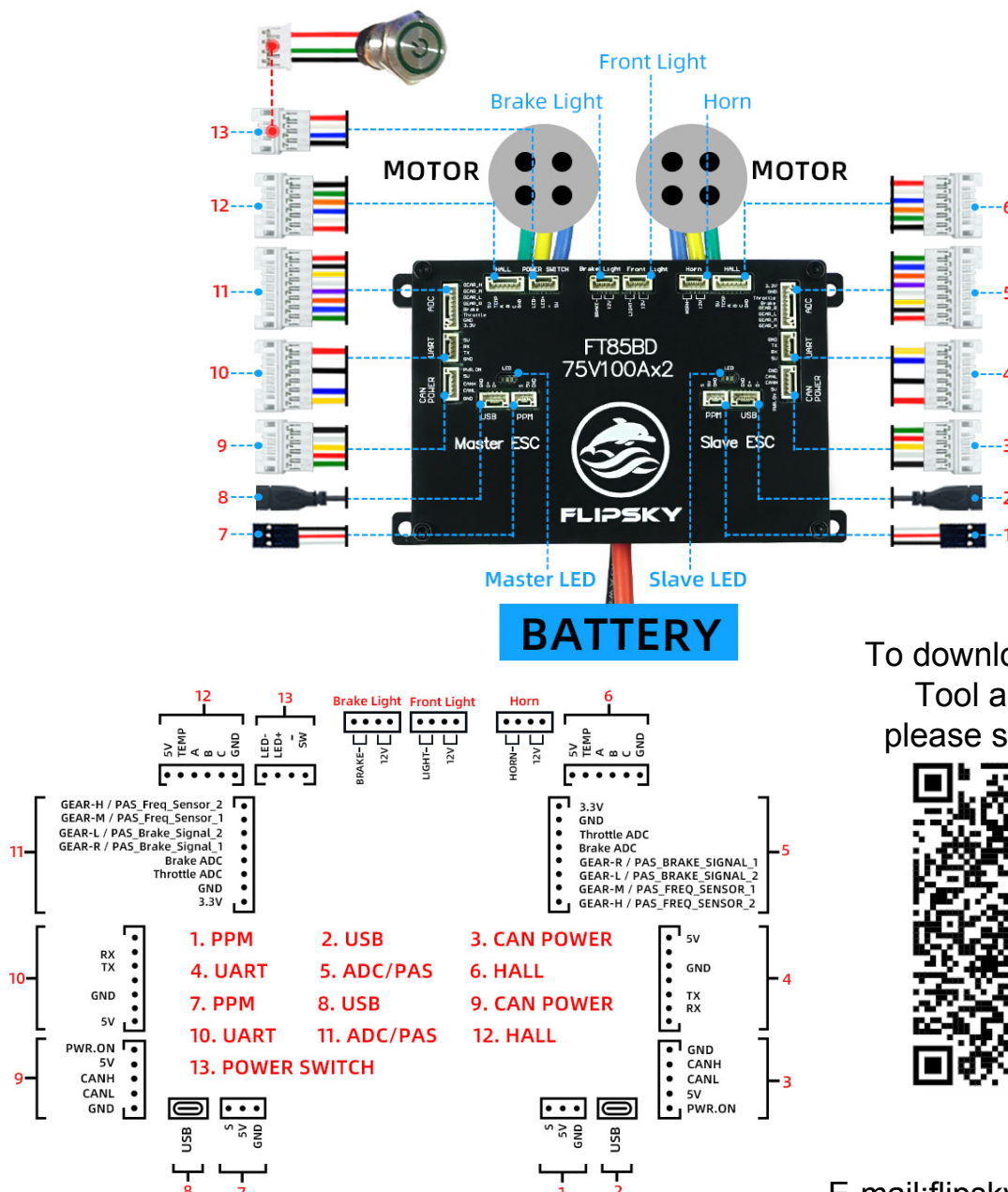
- Adopting ARM Cortex-M4 structure with the max 240MHz , improves PWM switching frequency and the motor RPM.
- Programmable via Flipsky ESC Tool to be compatible with multi-types of motor, different battery voltages and many control ways.
- Programmable via Flipsky remote controller VX4 Pro(remote firmware version: VX4Pro_V1.1 and above).
- Supporting motor FOC with low working noise and smooth starting.
- Supporting Real-time data and fault alerts.
- Supporting adaptive variable PWM frequency control.
- Supporting sensor mode(HALL, ENCODER) and sensorless mode(IPDS, HFPI, OPEN).

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- Supporting control types: Current, Duty Cycle, Speed control, Position Control
- Supporting PAS
- Protection functions: low and high voltage protection, over-current protection, temperature abnormally protection, MOSFET over-temperature and over current protection, battery over discharge protecting, motor over-temperature protection.
- Integrated smart switch function: supports sliding power on and automatic shutdown function; automatic shutdown time is adjustable via Flipsky ESC Tool.
- Supporting extra interface like front lights, brake lights and horn fixtures (12V 3A).
- Supporting multiple typed of motor temperature sensors such as NTC-10K-25℃, NTC-10K-100℃, PT1000, PTC-1K-100℃, KTY and so on.

Wiring diagram:



To download the Flipsky ESC Tool and User Manual, please scan the QR code:



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E-mail:flipskytech@outlook.com

FT85BD规格书及产品特点（版本V1.5）

规格参数：

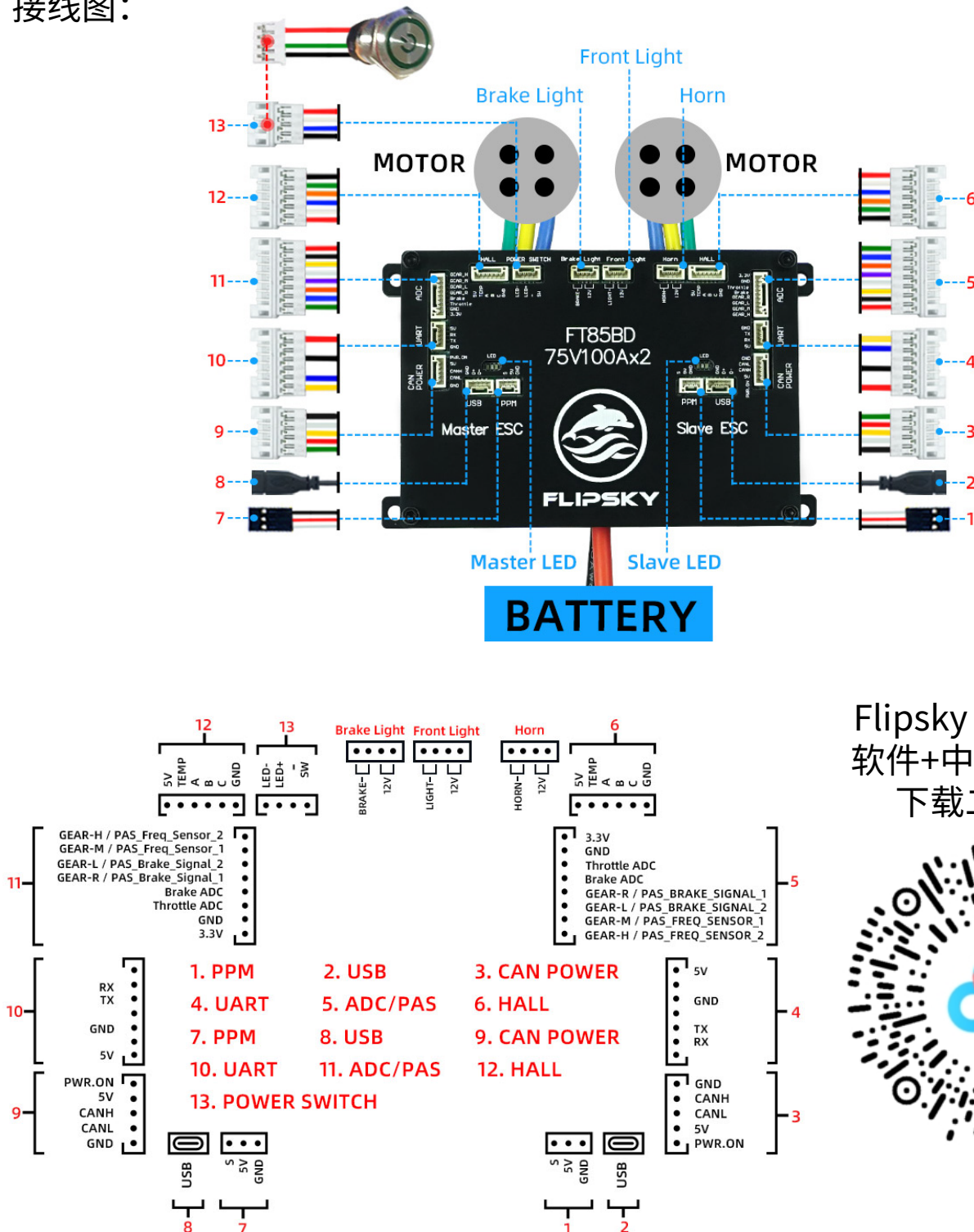
- 固件：V1.5及以上，支持通过Flipsky ESC Tool升级
- 电压：14V-85V (安全范围：4S-20S)
- 持续电流：100A/Single; 200A/Dual （取决于电调安装位置以及设备环境的温度、湿度）
- 峰值电流：250A/Single; 500A/Dual
- BEC:5V@1A
- 电力线缆：10 AWG
- 电机线缆：12 AWG
- 调参工具：Flipsky ESC TOOL
- PWM变频控制：YES
- 多挡位限速控制：YES(UART、ADC)
- 再生制动：YES
- 通讯端口：USB, CAN, UART, PPM, ADC
- 控制信号类型：PPM, ADC, PAS, UART （目前兼容 VX1、VX1Pro、VX2、VX3、VX3Pro、VX4Pro等遥控器）
- 启动模式：IPDS(无感轻载)，HFPI(无感重载)，HALL，ENCODER（ABI编码器，AS5047绝对式编码器SPI接口），OPEN(无感开环启动)
- 牵引力控制：YES
- 速度档位：低速档，中速档，高速档，倒车档
- 集成智能开关功能：支持滑动开机、支持自动关机功能、支持修改自动关机时间（待机电流低至50uA）
- 自复位开关带绿色LED按钮直径：16mm
- 尺寸：长132mm（带耳朵）、长116mm（不带耳朵）、宽81mm、高26.5 mm（带外壳）

产品特点：

- 采用ARM Cortex-M4架构，高达240MHz主频，可提高PWM控制频率和电机转速上限
- 可通过 Flipsky ESC Tool 编程，兼容多种电机、不同电池电压和多种控制方式；
- 支持 Flipsky VX4Pro 遥控器调参（固件版本：VX4Pro_V1.1 以上版本）；
- 支持 FOC，工作噪音低，启动平稳；
- 支持实时数据和故障报警支持自适应PWM变频控制
- 集成3组12V功率电源，支持前照灯、刹车灯、喇叭装置（12V电源最大输出电流3A）
- 支持PAS踏频传感器

- 支持传感器模式（HALL、ENCODER）和无传感器模式（IPDS、HFPI、OPEN）
- 支持控制类型：电流（Current），占空比（Duty Cycle），速度控制（Speed Control），位置控制（Position Control）
- 保护功能：低压保护；高压保护；电机过温保护；MOS管过流保护；电池过放保护；MOS管过温保护
- 支持多种类型电机温度传感器类型，包括NTC-10K-25°C、NTC-10K-100°C、PT1000、PTC-1K-100°C、KTY系列等多种类型温度传感器

接线图：



Flipsky ESC TOOL
软件+中文用户手册
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